

A photograph of a modern university building at night, with its interior lights glowing and reflecting in a body of water in the foreground. The building has a mix of brick and glass facades. The sky is a deep purple.

AGILE SOFTWARE ENGINEERING:

PRODUCT PLANNING AND REFINEMENT

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Last lecture objectives

An overview of eXtreme Programming (XP)

Basic skills in estimation of software development tasks

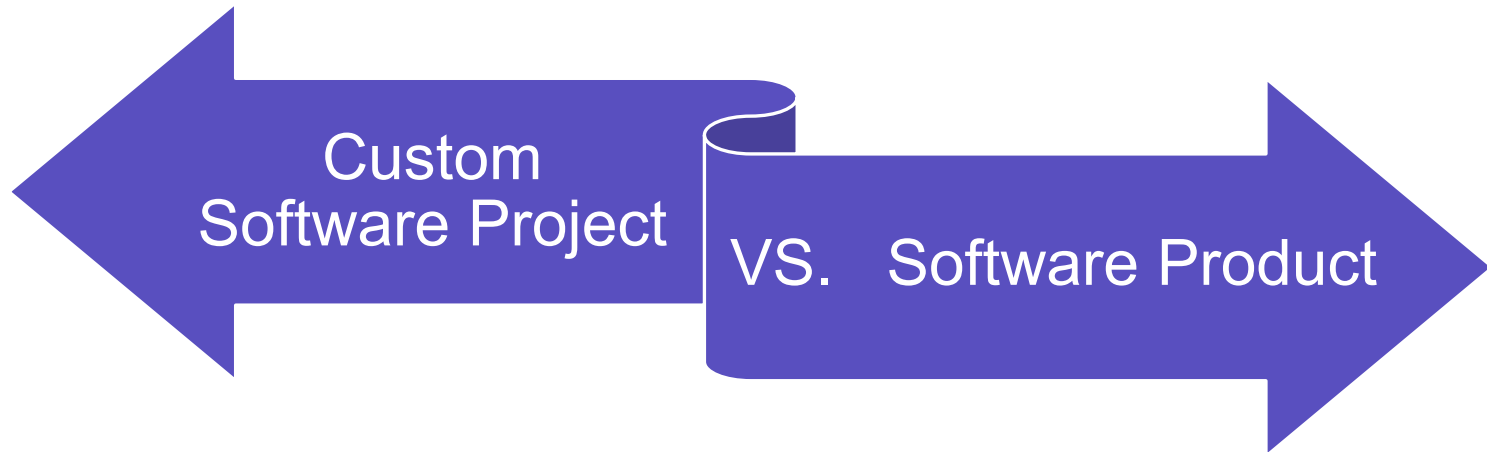
Prepared to chose and adapt a development method for a specific project

Lecture objectives

An overview of product planning and refinement techniques.

Skills in working with product visions, roadmaps, user story mapping, the product backlog, and product planning.

Competencies to manage the development of a software product.



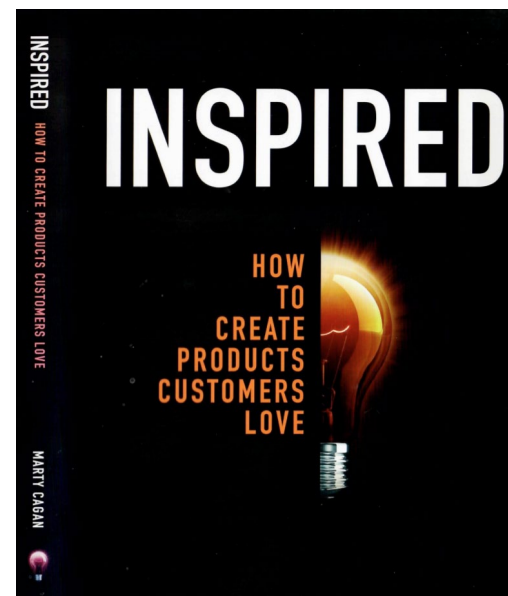
What is the difference between a project versus a product in:

1. How many times is it delivered?
2. What is most in focus: verifying all requirements are implemented or validating that they are fit for use?
3. Which of the key activities in software engineering (Specification, design and development, validation, evolution) are considered?

Product Planning

A custom software project is unique $A \rightarrow B$

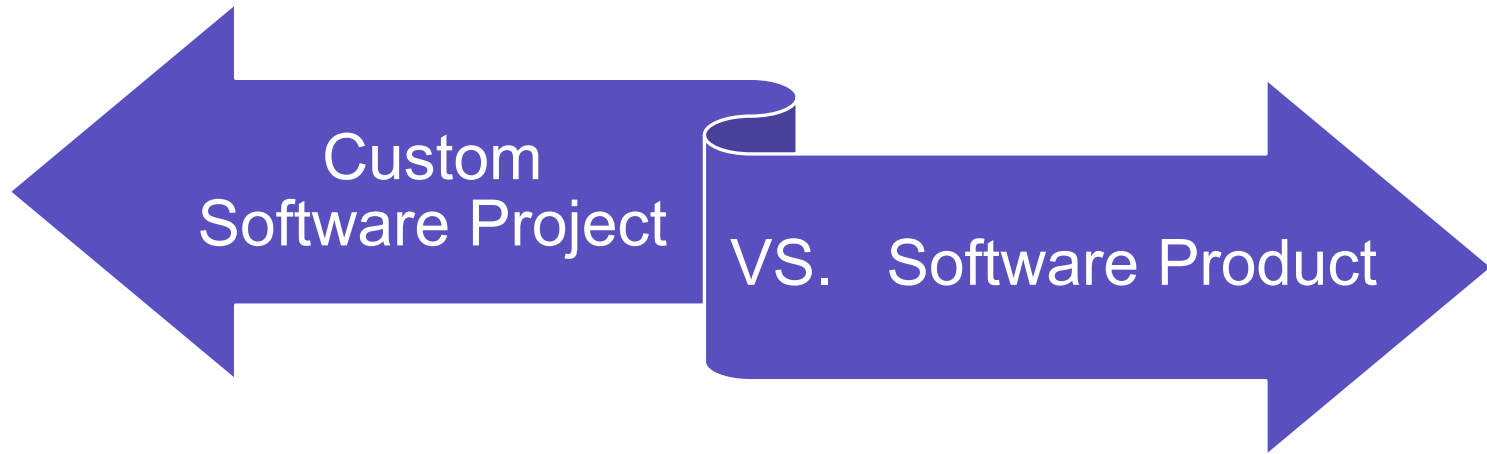
A product demands ongoing development $A \rightarrow \infty$



Succeeding with agile methods in software **product** teams:

1. Product manager is product owner
2. Product planning is shorter and rolling
3. Do design as part of backlog refinement
4. Design in small and independent chunks
5. Replace requirements docs with prototypes & user stories
6. Let the team chunk functionality into sprints
7. Socialize and communicate between team, PO and others
8. Product increments are released to users when sufficient functionality is available
9. Demo current state of product every sprint
10. Train people in agile

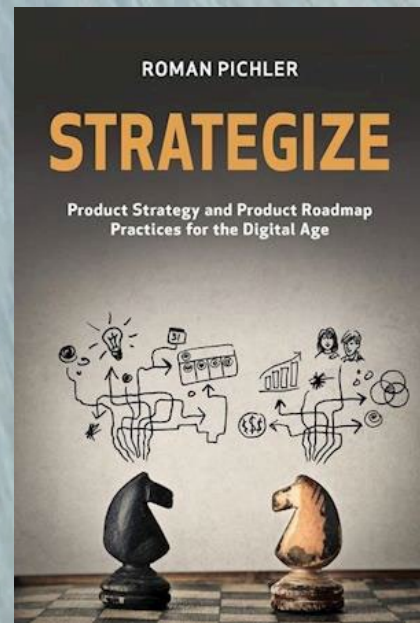




What is the difference between a project versus a product in how requirements are established from the understanding of needs and users:

1. When are requirements established?
2. How often are requirements adjusted?
3. What drives changes to requirements?



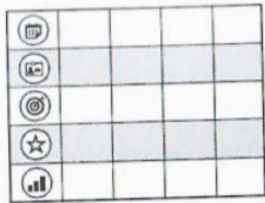


PRODUCT VISION AND ROADMAP (PICHLER)

Product roadmaps



PRODUCT ROADMAP



Strategic plan; describes how the product is likely to grow across several product releases.



PRODUCT BACKLOG

Tactical tool; describes the details, including epics and user stories, that have to be implemented to create one or more releases.

FIGURE 39: Product Roadmap and Product Backlog

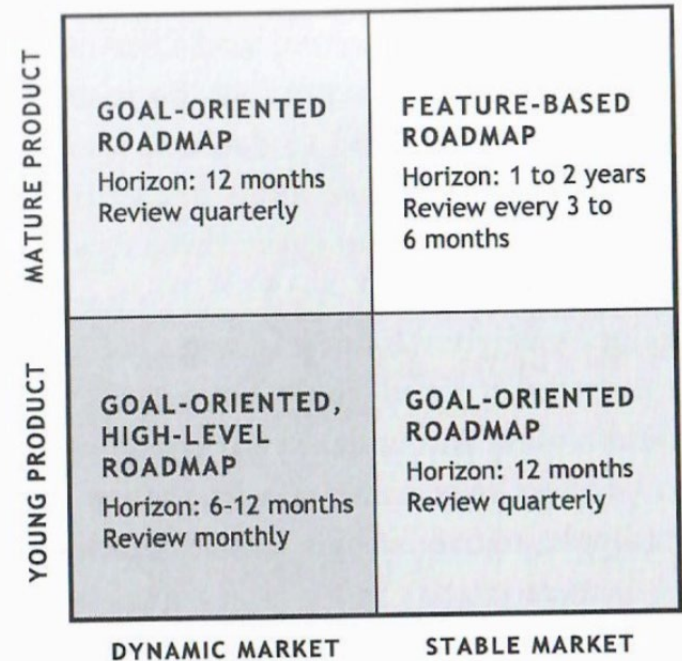


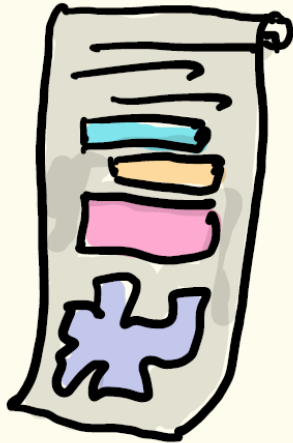
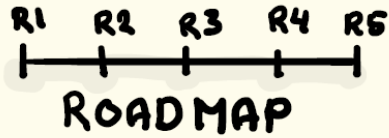
FIGURE 37: Roadmap Selection Matrix

PRODUCT VISION

STRATEGY



VISION



PRODUCT BACKLOG



VISION

Your overarching goal for creating the product



TARGET GROUP

Market
Customers
Users



NEEDS

Problem
Solved
Benefit
Provided



PRODUCT

What is it
How is it
special
Can it be
created



BUSINESS GOAL

Company
benefits



Elevator Pitch

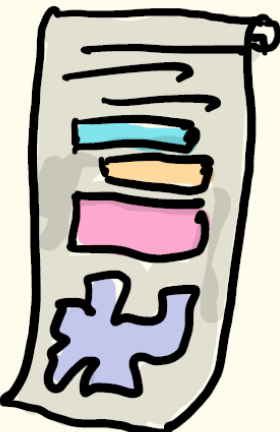
*For <target customer/user>
who <statement of need>
the <product name>
is a <product category>
that provides these <key benefits>
unlike <primary competition>
our product <is different in these ways>*



PRODUCT ROADMAP


VISION

R1 R2 R3 R4 R5
ROADMAP


PRODUCT BACKLOG

	RELEASE DATES OR TIME FRAMES
	THE RELEASE NAMES
	THE REASON FOR CREATING THE NEW RELEASE
	THE HIGH LEVEL FEATURES NECESSARY TO MEET THE GOALS
	THE METRICS TO DETERMINE GOALS HAVE BEEN MET





PRODUCT BACKLOG, SPRINT PLANNING, USER STORY MAPPING

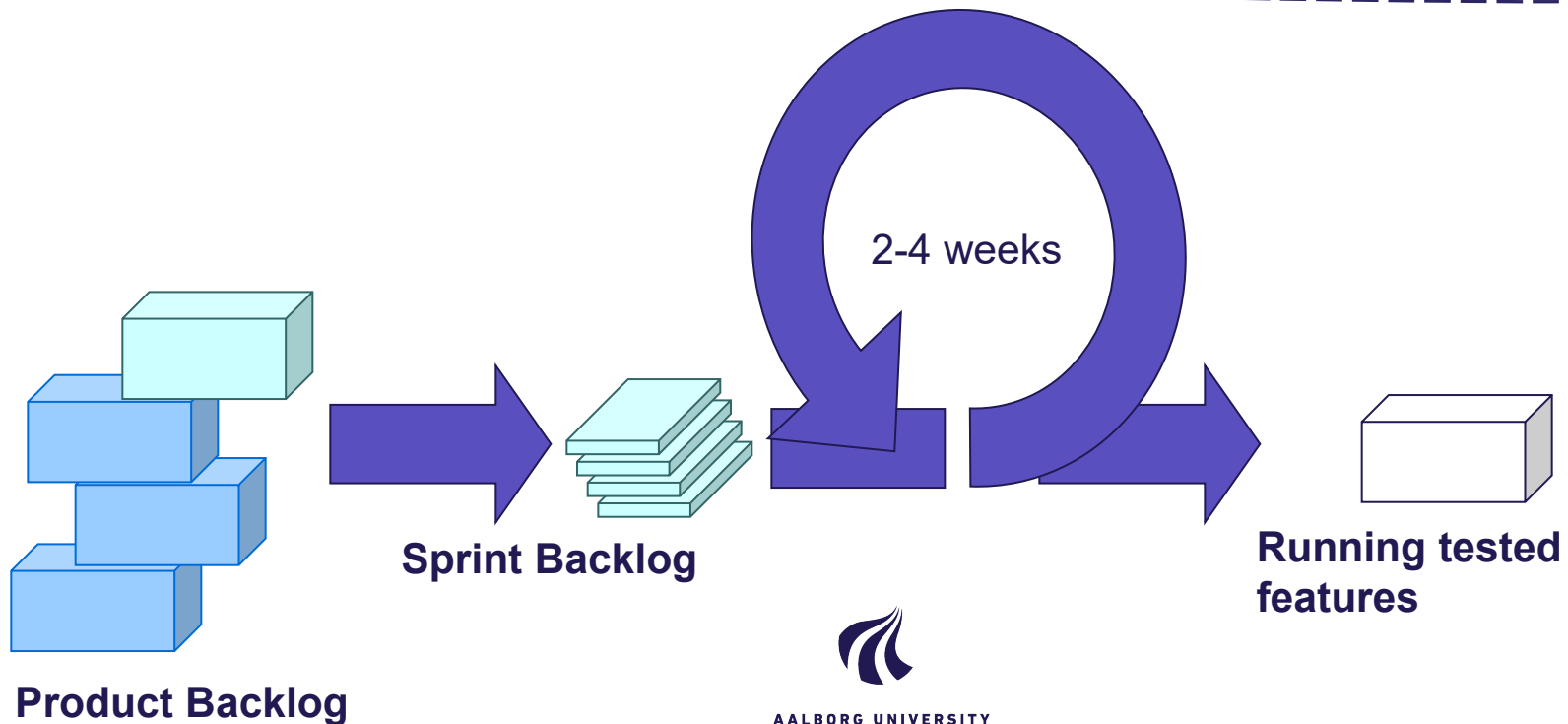
User Story delivery

User Role (*Who?*)

Desired Function (*What?*)

End Result (*Why?*)

As a **<type of user>**, I can **<immediate goal>** so that **<reason>**.



Personas (Roman Pichler)

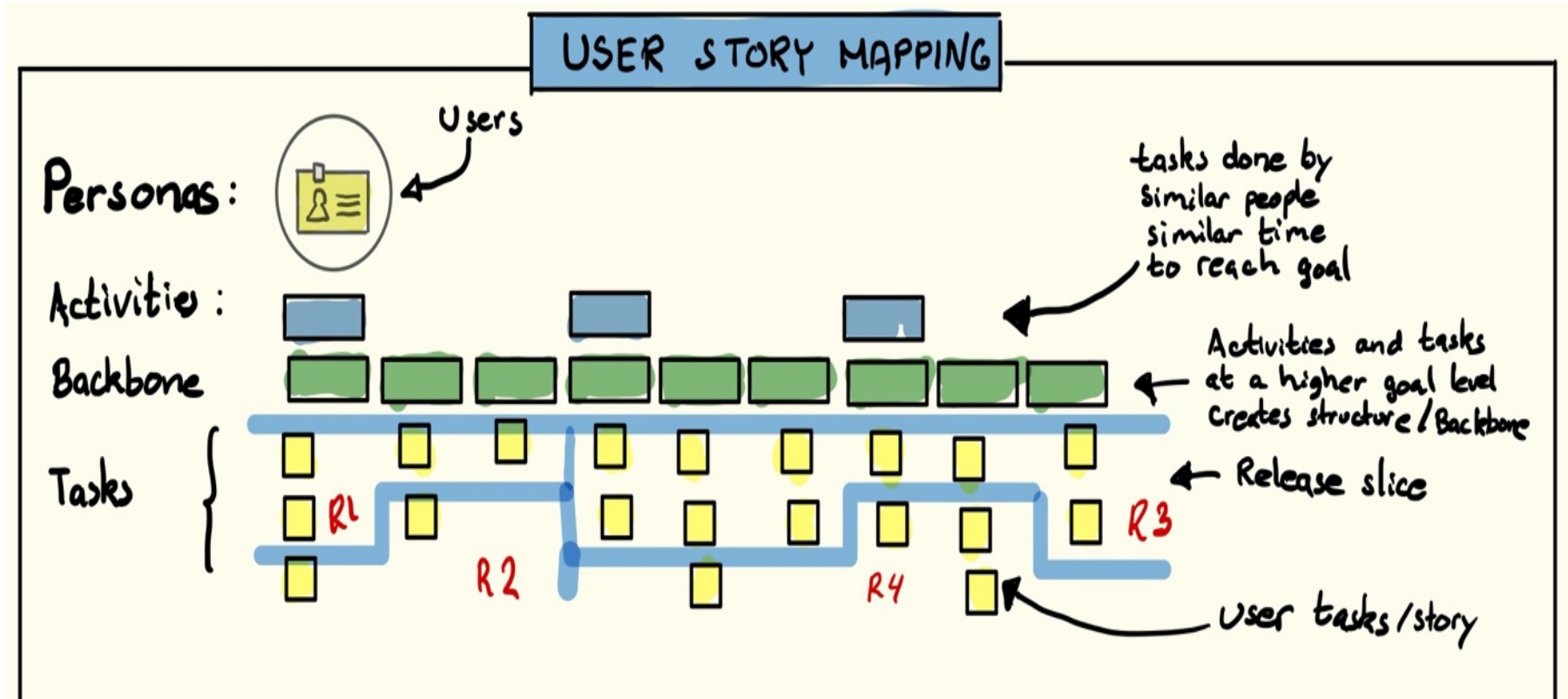
 PICTURE+NAME	 DETAILS	 GOAL
What does the persona look like? What is its name?	What are the persona's relevant characteristics and behaviors?	Why would the persona want to use or buy the product? What benefit should be achieved? Which problem should be solved?



 PICTURE+NAME	 DETAILS	 GOAL
 John	- Father of two - 44 y/o - Busy	- Healthy kids - Easy bribes

		Type of change		
		Do new things	Do things better	Stop doing things
Type of benefit valuation	Financial			
	Measurable			
	Observable			

User Story Mapping



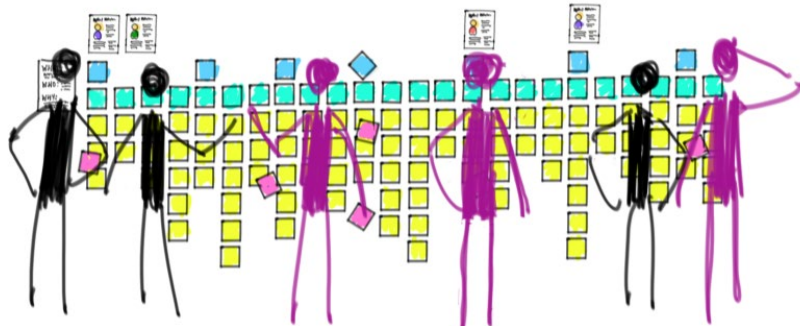
Story Map Process



1. Frame: The big story (What, Who, Why)



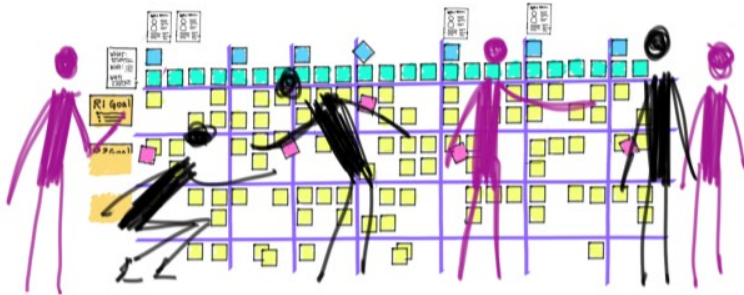
2. Map the big picture: The activities and high-level user tasks forming the backbone



3. Explore: Breakdown into subtasks



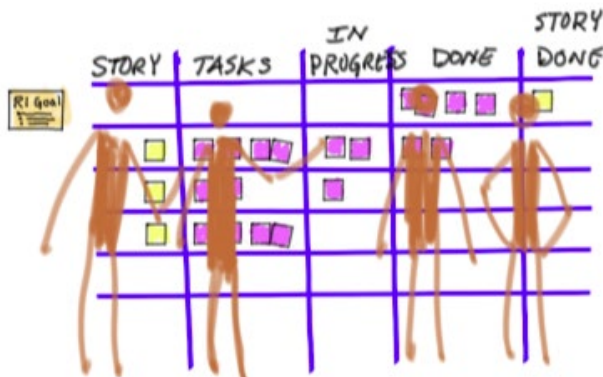
Story Map Process



4. Slice out Viable Releases: Increments with named outcomes and success metrics



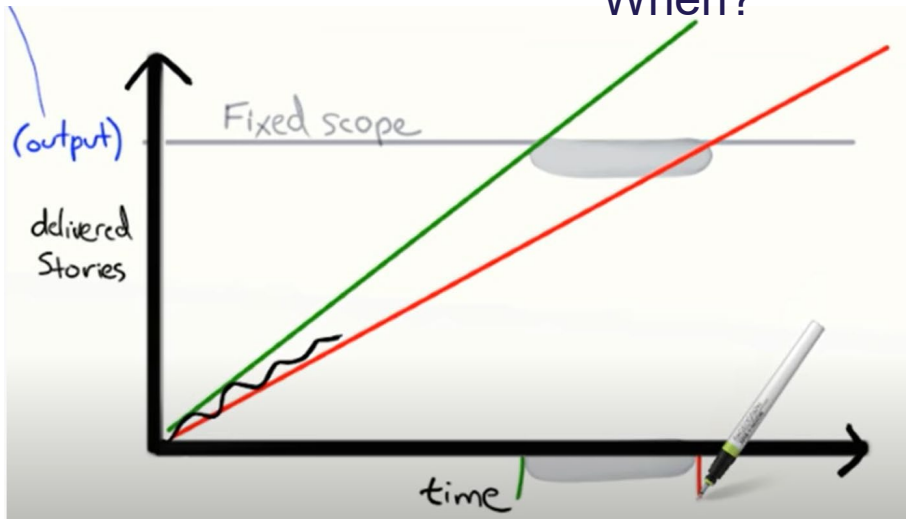
5. Slice out a Development Strategy: An opening, mid, and endgame for the product



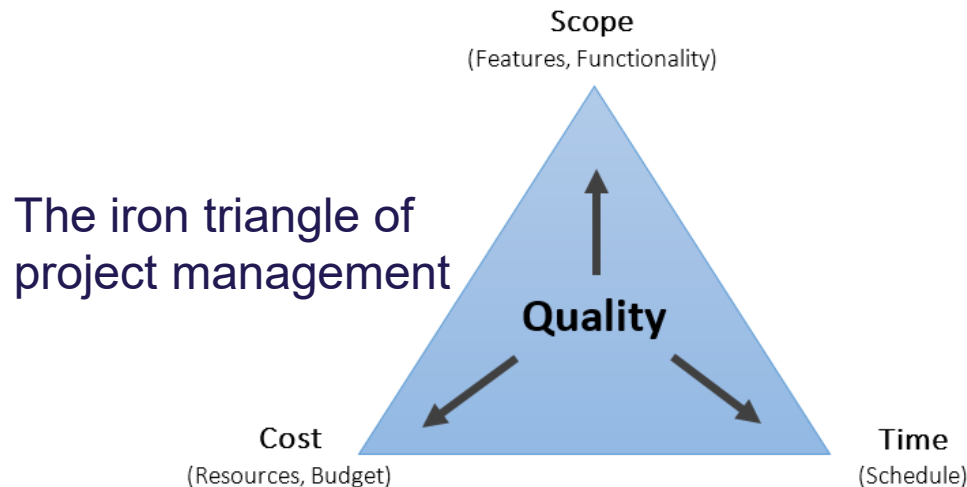
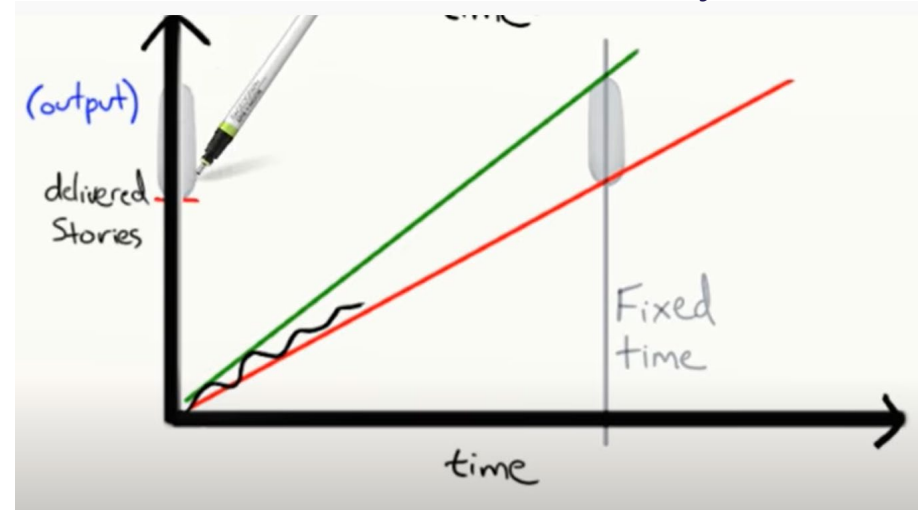
Forecasting User Story Delivery (Fixed Scope or Time)

(Video: Agile Product Ownership in a Nutshell)

When?



How much by?



Group exercises today

Exercise 1: Establish a Product Vision either for your semester project or an imaginary product (30%)

Exercise 2: Establish an elevator pitch (20%)

Exercise 3: How is your product launched? (20%)

Exercise 4: Brainstorm possible users and stakeholders of your product (10%)

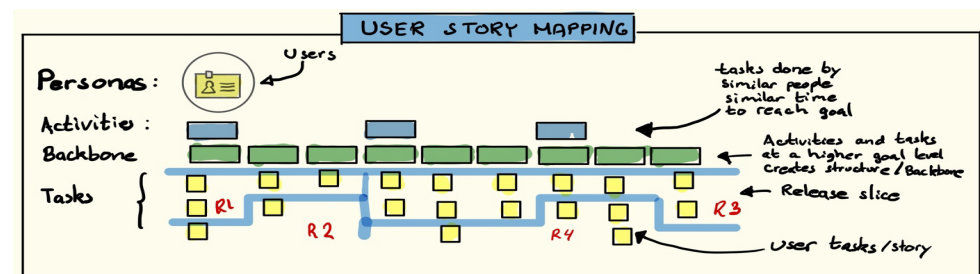
Exercise 5: Use the template and create 1-3 Personas (10%)

Exercise 6: Establish a user story map for your product (20%)

(110% 😊)

 VISION Your overarching goal for creating the product			
 TARGET GROUP Market Customers Users	 NEEDS Problem solved Benefit Provided	 PRODUCT What is it How is it special Can it be created	 BUSINESS GOAL Company benefits

	RELEASE DATES OR TIME FRAMES			
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